Sawyer Knowles

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Professional Experience

SS&C Advent ➤ ***Software Engineer, Full Stack – Advent Market Data*** ***2024 – Present***

- Developed client-facing features using Angular, TypeScript, and .NET REST APIs, expanding product capabilities and improving internal support workflows.
- Architected and implemented an extensible auditing framework capturing UI and API activity to optimize troubleshooting, compliance, and client support.
- Improved homepage load performance by 80% through redesigning MongoDB aggregation pipelines, query execution, and indexing strategy.
- Integrated Advent Market Data with SS&C's internal authentication and authorization platform, simplifying secure API access for enterprise clients.
- Drove feature development from technical design through deployment, collaborating with product managers and UX designers to translate requirements into production-ready solutions.
- Led a cross-functional hackathon team to prototype a generative AI feature that transformed natural language prompts into interactive market data dashboards and visualizations.

Sigma Connectivity, Inc. ➤ ***Software Engineer, XR Prototyping – Meta Reality Labs*** ***2022 - 2023***

Client: Meta Platforms, Inc. – Facebook Portal Display

- Developed a multi-user XR prototype connecting VR headset users with glasses-free 3D display users in a shared virtual environment, featuring real-time face and hand tracking, avatar animation, and inverse kinematics using Unity, ARKit, and Oculus XR to support immersive user interaction.
- Engineered virtual and physical multi-camera video systems to capture, align, and optimize autostereoscopic video using Python, OpenCV, and FFmpeg for 3D displays.
- Collaborated with software, electrical, and optical engineers on a small, fast-paced R&D team to rapidly develop and validate prototypes for emerging display technologies.

iD Tech ➤ ***Instructor, Game Development – Computer Science Academy*** ***2021***

- Taught advanced courses in C#, Unity, game development, and Agile software development.
- Mentored students in the design, implementation, debugging, code review, and presentation of software projects.

Projects

Wedding Website ➤ ***Full-Stack Web Application*** ***2025 – 2026***

- Designed and deployed my full-stack wedding website using Angular and Google Apps Script, integrating Google Sheets as a lightweight backend for interactive day-of event experiences.

Cards of the Ancient Zoo ➤ ***Narrative Deck-Building RPG*** ***2022***

- Led an Agile team through sprint planning, task coordination, and milestone delivery during development.
- Designed data-driven gameplay systems including card management, enemy strategy, and UI architecture using C# and Unity.

Space Escape VR ➤ ***Virtual Reality Escape Room Puzzle Game*** ***2022***

- Developed modular gameplay systems, UI, and scene management using Unreal Engine for a VR escape room experience.

Education

Michigan Technological University ***2018 – 2022***

Bachelor of Science, Computer Science • 3.6 GPA

Skills

- Languages: C#, TypeScript, Python, Kotlin, Java, SQL, C++, C
- Frontend: Angular, React, HTML, CSS
- Backend & Databases: .NET, REST APIs, MongoDB
- Developer Tools : Git, Jenkins, Postman, Unity, Unreal Engine, FFmpeg, Copilot, Claude Code